

LGE Scar Quantification Using Foundation Models for Cardiac Disease Classification

Athira J. Jacob^{1,2}, Puneet Sharma¹, and Daniel Rueckert^{2,3}

¹ Digital Technology and Innovation, Siemens Healthineers, Princeton, NJ, USA

² AI in Healthcare and Medicine, Klinikum rechts der Isar, Technical University of Munich, Germany

³ Department of Computing, Imperial College London, UK

Abstract. Late Gadolinium Enhancement (LGE) cardiac MRI (CMRI) is the gold standard for assessing myocardial viability, but myocardial scar quantification is challenging due to inter-observer variability. While deep learning methods can achieve high agreement within a dataset, their generalization and clinical utility remain underexplored. In this work, we develop an LGE scar quantification model trained on pseudo-ground truth masks from expert-generated segment-wise labels. The dataset includes 159 and 53 patients for training and testing, respectively. To enhance performance, the model is initialized with a CMRI foundation model pretrained on 36 million images. The model obtained an AUC of 0.96 and accuracy of 0.89 for LGE detection. The trained model is then applied to an independent dataset of 662 patients from a different institution, without any modification. Here, we demonstrate the utility of the calculated scar burden as a prognosticator for detecting myopathologies (normal, dilated, hypertrophic, ischemic), over and above other cine-based measurements, by improving AUCs by 1-6% for various disease classes. This study demonstrates that a scar segmentation model trained without expert-annotated pixel-wise ground truth masks can generate clinically meaningful scar burden measurements, offering valuable diagnostic insights.

Keywords: LGE scar segmentation · foundation model · cardiac disease classification · vision transformers

1 Introduction

Late Gadolinium Enhancement (LGE) imaging is the gold standard for assessing myocardial viability, providing clinical insights into scarred and fibrotic tissue. [11]. However, enhancement/scar analysis is challenging due to complex enhancement patterns, imaging artifacts (e.g. due to motion), scanner variations, and inter-observer inconsistencies in scar annotation, making it difficult to establish a precise and reliable pixel-wise ground truth (GT). On the other hand, binary labels indicating the presence or absence of enhancement are easier to create, though they are also subject to inter-user variability in borderline cases.

Traditional methods for LGE enhancement quantification rely on intensity thresholding techniques, such as the n-standard deviation (n-std) method and full-width at half-maximum (FWHM). While effective in some cases, these approaches are highly sensitive to imaging artifacts and variations in scan protocols [4, 7]. Deep learning (DL)-based methods have been explored to address these limitations; however, most prior studies train models on limited datasets with small numbers of manually annotated LGE images, restricting their generalizability [14].

This work presents a method for LGE quantification and demonstrates its utility in classifying myopathologies. The model is trained using "pseudo-ground truth" (GT) created from segment-wise scar labels. The proposed method obtains an AUC of 0.96 and an accuracy of 0.89 for LGE detection in the same-center validation. Additionally, we demonstrate that our predicted scar burden (scar volume as a percentage of myocardial volume) improves myopathy classification in an independent external cohort, improving AUCs by 1-6% and accuracy by 3%, without requiring additional fine-tuning. Our results demonstrate a scalable LGE segmentation framework that enhances both segmentation reliability and clinical decision-making.

2 Related Work

LGE Detection. No consensus exists on the optimal quantitative method for assessing infarction in LGE images. Commonly used approaches include Full Width Half Maximum (FWHM) [8, 1], and n-standard deviation (n-std) from remote myocardium. However, comparison studies [4, 7] reveal substantial inter-technique variability, highlighting concerns regarding reliability and reproducibility. Numerous studies have investigated the use of DL techniques for scar detection from LGE [20, 12, 5, 19]. Several of these, including [20], [5] and [19], were conducted on the publicly available EMIDEC challenge dataset [13]. This dataset comprises LGE-MRI scans for 150 patients, approximately two-thirds of whom have pathological findings. Various methods achieved a normal/abnormal classification accuracy of up to 91% on the test set of 50 patients [14]. While challenge datasets [13, 21] have been widely utilized, they present limitations that may hinder clinical translation, such as standardized image quality and the exclusion of slices without myocardium, which do not reflect real-world clinical data. Popescu et al [17] used 155 LGE-MRI datasets, and 246 synthetic LGE datasets to train a model to segment scar and obtain scar burden, and achieved a balanced accuracy of 75%. In [12], segmental information from diagnostic reports was used to learn presence/absence of LGE from LGE images. The method was developed on a cohort of 170 patients without any manual scar annotations (similar to our method), and reported an AUC of 0.875 for scar detection.

Disease Classification. Many studies have explored disease detection on the publicly available ACDC dataset [2] consisting of 150 patients, and achieved high accuracies of up to 0.96. However, this dataset did not include any LGE

images and patients with ambiguous characteristics were excluded, potentially affecting performance. Paciorek et al.[16] train a DL model to distinguish normal/abnormal patients with 88% accuracy in a test set of 40 patients, however the specific types of diseases were not distinguished. In [10], the authors trained a model to classify patients into 4 groups (normal and 4 disease classes) based on cine images and obtained AUCs of 0.86 to 0.95 for the various classes.

3 Methods

3.1 Data

Center 1: Algorithm Development Dataset. The dataset consisted of 212 patients, divided into 159 patients (1516 images) for training and 53 patients (497 images) for testing, with 20% of the training set used for validation. Scans were acquired on 1.5T Siemens MAGNETOM Aera scanners using T1-weighted inversion-recovery sequences (TR/TE: 3.4/1.3 ms; slice thickness: 6 - 7 mm; in-plane resolution: 1.3 to 2.5 mm²). Segment-wise binary scar labels (based on the AHA 17-segment model) were annotated by an expert reader (1 = scar present, 0 = no scar), giving 17 labels for each patient. LGE enhancement was observed in 62% of patients.

Center 2: Disease Classification Dataset. The dataset was an independent cohort of 662 patients (different continent and healthcare system). Scans were acquired on 1.5T Siemens MAGNETOM Avanto scanners using T1-weighted PSIR sequences (TR/TE: 2.4 /1.1ms; slice thickness: 8 mm; in-plane resolution: 1.3 to 2.6 mm²), along with cine bSSFP in standard short and long views. Patients were classified into four groups: a) normal (NORM, 43%): no structural or functional abnormalities on cardiac MRI, b) dilated cardiomyopathy (DCM, 11%): left ventricular (LV) dilation and systolic dysfunction, c) hypertrophic cardiomyopathy (HCM, 13%): LV hypertrophy without abnormal loading conditions and d) ischemic heart disease (IHD, 33%): myocardial ischemia from coronary artery disease (CAD), often leading to infarction and scarring detectable on LGE imaging. The labels were assigned based on expert consensus review of images, reports, and records by two experienced cardiovascular radiologists.

3.2 Baseline Methods

We implemented the n-standard deviation (n-std) thresholding algorithm, defining scar as myocardial pixels with intensity above $\mu + n\sigma$, where μ and σ were computed over normal (non-scarred) myocardium. First, The myocardium was automatically segmented and divided into AHA segments using the anterior right ventricular insertion point (RVIP) [3]. For this, we used previously trained deep learning models for both myocardium segmentation and RVIP detection, offering a fast and reliable alternative to manual annotation. On independent test sets, these models achieved a Dice score of 0.89 for myocardium segmentation

and a 3.1 mm localization error for RVIP detection—within the range of inter-observer variability [22]. To estimate an upper bound on performance, normal myocardium regions were defined using expert-provided segment-wise labels (see Section 3.1, Center 1). Thresholds of $n = 2, 3$, and 4 were tested, with higher values yielding more conservative scar segmentation.

3.3 Psuedo-GT generation on Algorithm Development Dataset

The proposed model was trained using pseudo-ground truth (pseudo-GT) masks, derived from the n -std thresholding method in combination with segment-wise clinical labels (Section 3.1). First, n -std method was run with $n = 1.5$ for maximum scar coverage. This method often overestimated scar, producing false positives (FP) in normal segments (Fig. 1a–d). To mitigate this, we manually refined the masks by removing FP clusters within clinically labeled normal segments, while preserving clusters in positive segments (shown in Fig. 1a–d). Specifically, connected components of the mask that were FP were deleted, while TPs were left as-is without any modifications. This approach retained the consistency and sensitivity of intensity-based delineation while improving specificity. We note that this method can also be potentially automated.

3.4 Proposed Model

The proposed model utilizes a U-Net transformer architecture [6], with a ViTS/8 [18] encoder. The encoder was initialized with a CMR foundation model pre-trained in a self-supervised manner on 36 million cardiac MR images [9]. Ablation studies (Section 4.3) explore the effect of encoder architecture and pretraining. Input images were preprocessed by resampling to $1\text{mm} \times 1\text{mm}$ and center cropped to 224×224 pixels. The model was trained using the Jaccard loss and a cosine annealing learning rate schedule (initial LR: 0.001), with geometric (rotation, translation) and elastic deformations applied for data augmentation. A multi-task learning strategy was employed, simultaneously predicting both the scar and myocardium masks, as multi-channel outputs for each image. Scar supervision was provided by the pseudo-GT masks described in Section 3.3, derived from segment-wise labels and intensity thresholding. Myocardium masks were obtained from the pretrained segmentation model detailed in Section 3.2. Final predictions were generated by ensembling three models trained with different random seeds.

Post-processing. Predicted masks were postprocessed as follows: a) Small isolated clusters—defined as connected components with area $< 20 \text{ mm}^2$ and no other detected scar within a 5mm radius—were removed; (b) Slices above the basal extent of the left ventricle (LV) were excluded to avoid false positives near the LV outflow tract. The basal-most slice was identified using a pre-trained deep learning classifier applied to the corresponding cine stack. This location was then mapped to the LGE stack using slice position and orientation from the DICOM metadata. The full post-processing pipeline is illustrated in Fig. 1e.

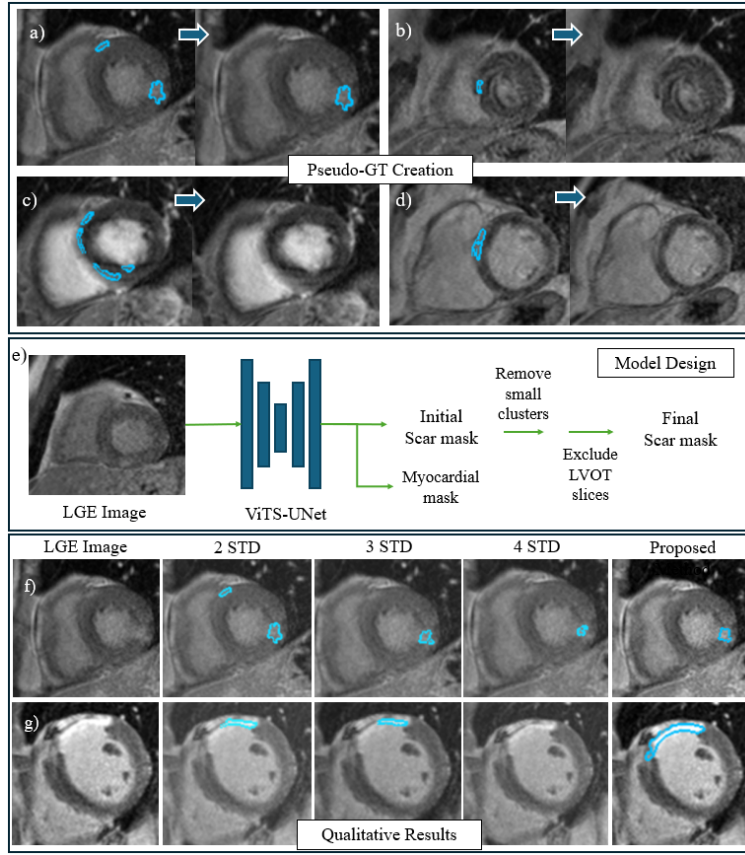


Fig. 1. (a–d) illustrate examples of initial masks from 1.5 std method that were refined to generate the psuedo-GT masks for training. False positive clusters were removed, while true positive clusters were kept as-is. (e) shows the pipeline used for inference. (f,g) Representative results on two patients from the same-center testing set.

3.5 Validation

Patient-wise LGE detection (Same-center). Performance was evaluated using patient-level metrics for direct comparison with clinical ground truth, including accuracy, balanced accuracy, AUC, sensitivity, specificity, and a confusion matrix. AUC was assessed using scar burden, while patients with nonzero scar burden were classified as LGE-positive for other metrics.

Disease Classification (External). For each patient, LGE magnitude images were selected based on DICOM metadata. The dataset included two acquisition types: single-shot (acquired in a single cardiac cycle, without breath-hold) and segmented (acquired over multiple cycles, with breath-hold). Segmented images

were preferred, when available, due to their higher spatial resolution for scar assessment. The trained scar quantification algorithm was applied without fine-tuning to estimate scar burden. This value was incorporated as an additional feature into an existing cine-based disease classification framework [10], which included features such as left ventricular ejection fraction, volumes, and strain. Two XGBoost classifiers were trained: (a) a cine-only model using 96 cine-image derived features and (b) a cine+LGE model using the quantified scar burden as an additional feature, giving 97 total features. The dataset was split into training(477), validation(54), and testing(171) sets, with 10-fold cross-validation. Performance was assessed using per-class AUC, weighted AUC, and overall accuracy on the test set. Feature importances are analyzed using SHAP analyses [15]. Note that the n-std algorithm, which requires a predefined normal myocardium region, could not be applied to this dataset due to the lack of such annotations.

Table 1. Results on same-center testing set (53 patients). The highest and second-highest values are highlighted in bold and underlined, respectively. Confusion matrices are presented as [True Negative, False Positive], [False Negative, True Positive].

Method	ROC-AUC	Accuracy	Balanced Accuracy	Confusion Matrix	Sensitivity	Specificity
2-std	0.76	0.68	0.61	$\begin{bmatrix} 6 & 15 \\ 2 & 30 \end{bmatrix}$	0.94	0.29
3-std	0.76	0.70	0.68	$\begin{bmatrix} 13 & 8 \\ 8 & 24 \end{bmatrix}$	0.75	0.62
4-std	0.71	0.66	0.70	$\begin{bmatrix} 19 & 2 \\ 16 & 16 \end{bmatrix}$	0.50	0.91
Proposed	0.96	0.89	<u>0.86</u>	$\begin{bmatrix} 16 & 6 \\ 0 & 31 \end{bmatrix}$	1.00	0.73

4 Results

4.1 Patient-wise LGE detection (Same-center)

The performance of the proposed method and the n-std algorithm ($n = 2, 3, 4$) on the 53 test patients is summarized in Table 1. Increasing n in the thresholding algorithm results in a trade-off between sensitivity and specificity, with lower n values yielding higher sensitivity but lower specificity, vice-versa. Regardless, n-std method was seen to be biased towards FP. Fig. 1 shows FPs due to RVIP (a), image artifacts (b), RV blood pool partial volume (PV) effects (c), and LVOT PV (d). The proposed model achieves the highest AUC and accuracy metrics. Representative scar segmentation results is shown in Fig. 1f-g.

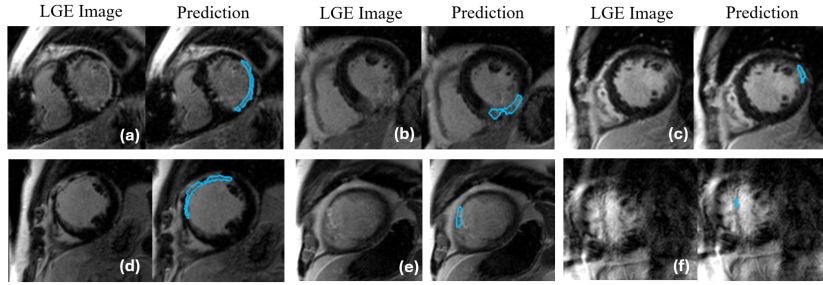


Fig. 2. Results on images from six patients on the external dataset. (a–d) True positive results, (e) Partial true positive, (f) False positive due to image artifact

4.2 Disease Classification (External)

The results for the disease classification dataset are shown in Table 2. Incorporating scar burden as a feature enhances classification performance across all classes, with AUC improvements of 1% for DCM, 2% for NORM and HCM, and 6% for IHD. The most significant gain is observed in IHD, aligning with its characteristic LGE presentation. SHAP analysis of the XGB model (Fig. 3) further emphasizes this, showing that LGE scar burden is the most influential feature for distinguishing IHD while also contributing to NORM and DCM classification, consistent with clinical protocols. Examples of predicted scar on this dataset are shown in Fig. 2.

Table 2. Results on disease classification in external dataset (n = 131 patients)

Model	ROC-AUC				Weighted AUC	Accuracy
	NORM	DCM	HCM	IHD		
Cine-only	0.95	0.87	0.93	0.83	0.90	0.76
Cine+LGE	0.97	0.88	0.95	0.89	0.93	0.79

4.3 Ablation Study - Encoder effect

The effect of encoder choice on performance was assessed by comparing three encoders: a) the CMR foundation model (ViTS) [9], b) a convolutional model trained on the related task of LGE myocardial segmentation (Dense121) (Section 3.2), and c) a ViTS model pre-trained on ImageNet. The CMR-trained ViTS achieved the best performance (Table 3), likely due to its ability to capture domain-specific cardiac features from large-scale data. The ImageNet-pretrained ViTS model achieved intermediate ROC-AUC but the lowest accuracy metrics, likely due to the domain gap. The convolutional encoder trained on the related task of myocardium segmentation achieved intermediate accuracy metrics but the lowest AUC, and may have been constrained by its task-specific training.

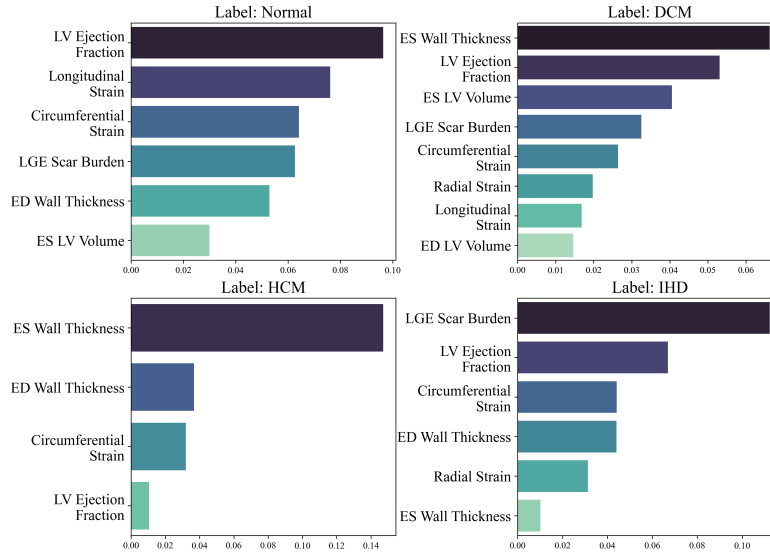


Fig. 3. Feature attributions for various disease classes using SHAP analyses.

Table 3. Ablation study: Effect of different encoders on performance. $n = 53$ patients

Encoder Type	ROC-AUC	Accuracy	Balanced Accuracy
CMR Foundation Model (ViTS/8)	0.96	0.89	0.86
Convolutional encoder (Dense121) pretrained on related task	0.82	0.81	0.83
ViTS/8 pretrained on ImageNet	0.89	0.75	0.71

5 Conclusions

This study presents a deep learning-based method for LGE scar quantification that demonstrates strong performance, cross-institution generalizability, and clinical utility in downstream disease classification tasks. Notably, the approach does not rely on expert-delineated pixel-wise ground truth masks. However, limitations include the use of single-vendor datasets, the absence of expert-annotated LGE masks for model comparison, and the lack of LGE ground truth labels in the disease classification cohort. Despite these constraints, the derived scar burden proved clinically meaningful, improving disease detection metrics and contributing to model interpretability through feature attribution analyses. Overall, the proposed method offers a scalable and annotation-efficient solution for generating robust LGE scar burden estimates, mitigating the challenges of manual labeling and inter-observer variability.

Disclosure of Interests. The authors AJJ and PS are employees of Siemens Healthineers.

Disclaimer. The concepts and information presented in this paper are based on research results that are not commercially available. Future commercial availability cannot be guaranteed.

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